## Example 3.36

There is a wooden piece of length 2*m*. A carpenter wants to cut it into two pieces such that the first piece is 40 *cm* smaller than twice the other piece. What is the length of the smaller piece ?

***Solution:***

Let us assume that the length of the first piece is $x$ *cm*.

Then the length of the second piece is (200*cm* – $x$ *cm*) i.e., (200 – $x$) *cm*.

According to the given statement (change *m* to *cm*),

First piece = 40 less than twice the second piece.

$$x = 2\times (200 - x) - 40$$

$$x = 400 - 2x - 40$$

$$x + 2x = 360$$

$$3x = 360$$

$$x = \frac{360}{3}$$

$$x = 120$$

**Think**



Suppose we take the second piece to be $x$ and the first piece to be (200 – $x$), how will the steps vary? Will the answer be different?

Hence,

(i) Thus the length of the first piece is 120 cm

(ii) The length of second piece is 200cm – 120cm = 80cm, which happens to be the smaller.

## Example 3.37

Mother is five times as old as her daughter. After 2 years, the mother will be four times as old as her daughter. What are their present ages?

***Solution:***

| Age / Person | Now | After 2 years |
|---|---|---|
| Daughter | $x$ | $x+2$ |
| Mother | $5x$ | $5x+2$ |

Given condition: After two years, Mother's age = 4 times of Daughter's age

$$5x + 2 = 4(x + 2)$$

$$5x + 2 = 4x + 8$$

$$5x - 4x = 8 - 2$$

$$x = 6$$

Hence daughter's present age = 6 years;

and mother's present age = $5x = 5 \times 6 = 30$ years

## Example 3.38

The denominator of a fraction is 3 more than its numerator. If 2 is added to the numerator and 9 is added to the denominator, the fraction becomes $\frac{5}{6}$. Find the original fraction.

***Solution:***

Let the original fraction be $\frac{x}{y}$.



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